

PROJECT TESTING



Performance Model Test

 DATE
17 7/9/2026

 NAME
K.Hemanth Kumar

 PROJECT
OptiCrop – Smart
Agricultural
Production
Optimization Engine

 MAX MARKS
5 Marks

Functional Test Scenarios

TEST ID	SCENARIO	TEST STEPS	EXPECTED RESULT	ACTUAL RESULT	RESULT
FT-01	User Input Validation (required fields, invalid values)	Enter valid and invalid soil/climate values in the input fields.	Valid inputs are accepted; invalid inputs show an error message.	Application accepted valid inputs and showed an error for invalid inputs.	Pass
FT-02	Crop Recommendation (predict suitable crop)	Enter complete soil/climate details and click "Recommend a crop".	System predicts the most suitable crop based on the input.	Prediction displayed successfully.	Pass
FT-03	Model Integration Check (ML model with Flask)	Submit soil/climate details to the application.	Application sends input to the ML model and receives a prediction.	Model returned a prediction successfully.	Pass
FT-04	Application Navigation (Home, About, Find your crop)	Navigate between all pages.	Pages load correctly without any errors.	All pages loaded successfully.	Pass

Performance Test Scenarios

TEST ID	SCENARIO	TEST STEPS	EXPECTED RESULT	ACTUAL RESULT	RESULT
PT-01	Response Time Test	Submit soil/climate details and measure response time.	Prediction should be displayed within 3 seconds.	Prediction displayed within 3 seconds.	Pass
PT-02	Multiple Prediction Test (repeated requests)	Perform multiple predictions one after another.	Application should respond correctly without slowing down.	Multiple predictions completed successfully without slowdown.	Pass
PT-03	Application Stability Test (long run, different inputs)	Run the application continuously with varied inputs.	Application should work without crashing or errors.	Application remained stable throughout testing.	Pass